

QING LI

Earth, Ocean and Atmospheric Sciences Thrust, Function Hub
The Hong Kong University of Science and Technology (Guangzhou)
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EDUCATION

- 2018 **Ph.D. Earth, Environmental and Planetary Sciences**,
Brown University, Providence, RI, USA
Advisor: B. Fox-Kemper
PhD thesis: *Langmuir Turbulence and Its Effects on Global Climate*
- 2013 **M. S. Meteorology**, Peking University, Beijing, China
Advisor: H. Yang
Master's thesis: *Numerical Simulations of a Fully Coupled Aqua-Planet:
Mean Climate and Meridional Heat Transport*
- 2010 **B. S. Atmospheric Sciences**, Peking University, Beijing, China
Senior thesis advisor: H. Yang
Senior thesis: *Lagrangian Analysis on the Circulation in the Pacific Ocean*
- 2010 **B. S. Double Major in Economics**, Peking University, Beijing, China

RESEARCH INTERESTS

Ocean boundary layer turbulence, Ocean surface waves, Ocean modeling, Climate sciences

PROFESSIONAL APPOINTMENTS

- 2023.10 - **Cross-Campus Faculty Affiliate**, The Hong Kong University of Science and Technology
- 2021.08 - **Assistant Professor**, The Hong Kong University of Science and Technology (Guangzhou) (HKUST-GZ)
Earth, Ocean and Atmospheric Sciences Thrust, Function Hub
- 2021.08 - 2023.06 **Affiliate Assistant Professor**, The Hong Kong University of Science and Technology
Dept. of Ocean Science, School of Science
- 2018.08 - 2021.05 **Postdoctoral Research Associate**, Los Alamos National Laboratory (LANL)
Fluid Dynamics and Solid Mechanics, Theoretical Division
- 2013.08 - 2018.05 **Research Assistant**, Brown University
Dept. of Earth, Environmental and Planetary Sciences
The Institute at Brown for Environment and Society (IBES)
- 2010.09 - 2013.07 **Research Assistant**, Peking University
Dept. of Atmospheric and Oceanic Sciences, School of Physics

AWARDS AND RESEARCH GRANTS

- 2027.01 - 2030.12 **General Fund of National Natural Science Foundation of China: Studying the dynamics and mixing effects of ocean surface boundary layer turbulence under the influence of submesoscale fronts**, CNY 490k, PI

- 2024.01 - 2028.12 **Hong Kong Research Grants Council Areas of Excellence Scheme:** *Modeling the ocean turbulent mixing under multi-process interaction in the Pearl River estuary and adjacent shelf in Study of the regional earth system for sustainable development under climate change in the Greater Bay Area*, HKD 87.147m, Co-I (HKD 1m to HKUST-GZ, PI: J. Gan)
- 2023.01 - 2025.12 **Young Scientists Fund (Category C) of National Natural Science Foundation of China:** *Langmuir turbulence in a diurnal cycle and its effects on turbulent mixing of momentum and tracers in the upper ocean*, CNY 300k, PI
- 2023 **Guangdong Provincial Talent Program**, Guangdong Provincial Department of Science and Technology
- 2022.04 - 2025.03 **Center for Ocean Research in Hong Kong and Macau Research Project:** *Modeling the ocean boundary layer turbulent mixing: from open oceans to coastal oceans*, HKD 400k, PI
- 2022.04 - 2024.03 **Center for Ocean Research in Hong Kong and Macau Research Project:** *Impact of wave-state dependent sea-surface flux on the regional climate of East Asia in climate system simulations*, HKD 400k, Co-I (PI: X. Shi)
- 2022 **Nansha District High-Level Leading Talent Program**, Talent Development Bureau of Nansha District
- 2018.10 - 2020.09 **LANL Institutional Computing Grant:** *Better understanding of the air-sea fluxes using atmosphere-ocean coupled large eddy simulation*, 7 Mcpuhr. + 40.9 TB storage, PI

FELLOWSHIP AWARDS AND TRAVEL GRANTS

- 2025.06 **Invited Speaker:** ICERM Hot Topics Workshop - Synthesizing Research On Ocean Surface Waves In The New Arctic, Providence, RI, USA, Full travel support
- 2024.06 **Discussion Leader:** Gordon Research Conference on Ocean Mixing: Understanding the Role of Ocean Mixing Across Scales on Climate, Ecosystems and Ocean Solutions to Societal Problems, South Hadley, MA, USA, Partial travel support
- 2019.08 **Visiting Scientist:** National Center for Atmospheric Research, Boulder, CO, USA, Full travel support
- 2018.10 **Invited Participant:** Physical Oceanography Dissertation Symposium (PODS) X, Kailua-Kona, HI, USA, Full travel support
- 2016.09 **Travel Grant:** CLIVAR Open Science Conference, Qingdao, Shandong, China, Full travel support
- 2016.05 **Travel Grant:** Liège Colloquium on Submesoscale Processes: Mechanisms, Implications and new Frontiers, Liège, Belgium, Partial travel support
- 2015.09 - 2016.08 **Fellowship:** IBES Graduate Student Fellowship at Brown University
- 2014.10 **Travel Grant:** Institute for Mathematics and its Applications (IMA) Workshop on Impact of Waves Along Coastlines, Minneapolis, MN, USA, Full travel support
- 2014.08 **Travel Grant:** The Community Earth System Model Tutorial, Boulder, CO, USA, Full travel support
- 2013.09 - 2014.08 **Fellowship:** First-Year Graduate Student Fellowship at Brown University

TEACHING EXPERIENCE

At HKUST-GZ [S: Spring, F: Fall]

- 2026 F **Lecturer**, UFUG1501 General Physics I, co-lectured with X. Chu
- 2024 S **Lecturer**, UFUG1106 Honors Calculus II
- Lecturer**, UFUG1103 Calculus II, co-lectured with G. Zhang and W. Wang
- 2023-25 F **Lecturer**, EOAS5000 Introduction to Oceanography, co-lectured with Q. Ji and L. Yu
- 2023-25 S **Lecturer**, EOAS5002 Atmosphere-Ocean Dynamics
- 2023 S **Lecturer**, EOAS5004 Earth System Modeling, co-lectured with Q. Yang and L. Yu

2022 F **Guest Lecturer**, EOAS6000B *Global Carbon Cycle and Climate Change*
 2022 F, 24 S **Guest Lecturer**, FUNH6800 *Function Hub Seminar*
 2022-23 S **Guest Lecturer**, FUNH5000 *Introduction to Function Hub for Sustainable Future*
 2021-23 F **Guest Lecturer**, FUNH6770 *Professional Development for Function Hub*
 2021 F **Lecturer**, EOAS6000A *Ocean Circulation, Carbon Cycle, Ecosystems, and Changing Climate*, co-lectured with Q. Ji and L. Yu

Elsewhere

2020 Summer **Student Mentor**, Los Alamos National Laboratory
Parallel Computing Summer Research Internship with L. Van Roekel and M. Turner
 2017 Spring **Guest Lecturer**, Brown University
Ocean Circulation and Climate under B. Fox-Kemper
 2016 Fall **Guest Lecturer**, Brown University
Mathematical Methods of Fluid and Solid Geophysics and Geology under B. Fox-Kemper
 2016 **Sheridan Teaching Certificate I**, Brown University

ACADEMIC ADVISING

Postdoc Scientists Mentor: Yaoru Pan (with B. Fox-Kemper, 2022-2023)
PhD Students Prime Advisor: Yunzhuo Zhang (2024-), Xinghao Jiang (2023-), Shihao Zou (2023-), Wentao Pan (2022-2026), Zheng Wei (2022-2026)
PhD Students Co-advisor: Mingming Zhu (with Z. Liu, 2025-), Hanshu Wang (with J. Chi, 2023-), Yaning Wang (with Z. Liu, 2023-), Zhouxiao Liu (with L. Yu, 2022-), Zhuowei Xu (with L. Yu, 2022-)
PhD Thesis Examiner: Xia Chai (advisor: Y. Wang, 2026), Weicong Cheng (advisor: J. Gan, 2024)
MPhil Students Advisor: Yankun Liu (2023-2025), Cheng Jin (2023-2024)

SERVICE TO HKUST-GZ

2026.09 - **Member:** HKUST-GZ Summer School Committee
 2023.12 **Member:** Acceptance expert team of the HKUST-GZ AI/HPC Data Center
 2023.04 - 2024.08 **PG Coordinator:** Earth, Ocean and Atmospheric Sciences (EOAS) Thrust
 2023.04 - 2024.08 **Liaison:** Red Bird MPhil (RBM) Program liaison at EOAS
 2021.09 - 2024.08 **Chair:** Postgraduate (PG) Program Committee at EOAS
 2021.09 - **Member:** PG Program Committee at EOAS
 2021.10 - 2022.07 **Member:** RBM Selection and Interview Committee

SERVICE TO THE PROFESSION AND ACADEMIC LITERATURE

2025.01 **Session Co-Chair:** with B. Chen and P. Wang, 7th Xiamen Symposium on Marine Environmental Sciences, Xiamen, Fujian, China. *Session: Ocean boundary layer turbulence: dynamics and its impact on the Earth system*
 2020.02 **Session Co-Chair:** with I. Savelyev, G. Wagner, and L. Johnson, Ocean Sciences Meeting, AGU/ASLO/TOS, San Diego, CA, USA. *Session: Turbulent mixing of the ocean surface boundary layer: Observation, Simulation, and Parameterization*
 2018.05 **Session Chair:** KITP Conference on Frontiers in Oceanic, Atmospheric, and Cryospheric Boundary Layers, Santa Barbara, CA, USA. *Session: Interdisciplinary*
 2015.08 **Student Volunteer:** Abstract sorting for 68th Annual Division of Fluid Dynamics Meeting, APS, Boston, MA, USA

Reviewer: National Science Foundation, National Natural Science Foundation of China
 Acta Oceanologica Sinica, Deep-Sea Research Part I: Oceanographic Research Papers, Deep-Sea Research Part II: Topical Studies in Oceanography, Geophysical Research Letters, Geoscientific Model Development, Journal of Advances in Modeling Earth Systems, Journal of Atmospheric and Oceanic Technology, Journal of Climate, Journal of Computational Physics, Journal of Geophysical Research: Atmospheres, Journal of Geophysical Research: Oceans, Journal of Physical Oceanography, Journal of Turbulence, Marine Geodesy, Nature Climate Change, Ocean Dynamics, Ocean Modelling, Ocean Science, Physics of Fluids, Science Advances, The Cryosphere

PUBLICATIONS

Underline indicates a student in the group, star indicates the corresponding author

- [A.1] S. Zou, D. Hauser, C. Peureux, Q. Li^{*}, Fusing SAR and SWIM observations to improve the global Stokes drift estimates, *Remote Sensing of Environment* 347 (2026) 115687. doi:10.1016/j.rse.2026.115687.
- [A.2] Q. Li^{*}, B. Chen, Turbulence in the ocean mixed layer, in: E. Bou-Zeid, S. Sarkar (Eds.), *Environmental Turbulence: From Fundamental Physics to Critical Applications*, Academic Press, 2026, pp. 307–333. doi:10.1016/B978-0-32-395873-8.00018-8.
- [A.3] X. Jiang, Q. Li^{*}, Impact of parameterized wind-driven boundary layer turbulence on the development and evolution of submesoscale mixed layer eddies in an idealized front, *Ocean Modelling* 202 (2026) 102741. doi:10.1016/j.ocemod.2026.102741.
- [A.4] Q. Li^{*}, Large eddy simulations of stabilizing effects induced by opposing Eulerian shear and Stokes drift shear in an idealized ocean surface boundary layer, *Journal of Physical Oceanography* 56 (4) (2026) 855–871. doi:10.1175/JPO-D-25-0077.1.
- [A.5] W. Pan, Q. Li^{*}, Transient response of Langmuir turbulence to abrupt onset of surface heating, *Physical Review Fluids* 11 (2) (2026) 024606. doi:10.1103/jjdt-zp9n.
- [A.6] Z. Wei, Q. Li^{*}, B. Chen, A direct assessment of Langmuir turbulence parameterizations in idealized coastal merging boundary layers, *Journal of Advances in Modeling Earth Systems* 18 (2026) e2025MS004993. doi:10.1029/2025MS004993.
- [A.7] J. Huang, M. Chamecki, Q. Li, B. Chen^{*}, The role of longitudinal alignment between surface and bottom forcing on the full-column turbulence mixing in the coastal ocean, *Ocean Modelling* 199 (2026) 102637. doi:10.1016/j.ocemod.2025.102637.
- [A.8] X. Shi^{*}, Q. Li, D. V. Lestari, S. Lin, H. Su, The effects of sea-state-dependent surface fluxes on CESM2 climate simulations, *Journal of Advances in Modeling Earth Systems* 17 (2025) e2025MS005284. doi:10.1029/2025MS005284.
- [A.9] X. Fan^{*}, B. Fox-Kemper, N. Suzuki, Q. Li, P. Marchesiello, F. Auclair, P. P. Sullivan, P. S. Hall, Comparison of the Coastal and Regional Ocean COMMunity Model (CROCO) and NCAR-LES in non-hydrostatic simulations, *Geoscientific Model Development* 17 (2024) 4095–4113. doi:10.5194/gmd-17-4095-2024.
- [A.10] A. Garanaik^{*}, F. Pereira, K. Smith, R. Robey, Q. Li, B. Pearson, L. Van Roekel, A new hybrid mass-flux/high-order turbulence closure for ocean vertical mixing, *Journal of Advances in Modeling Earth Systems* 16 (2024) e2023MS003846. doi:10.1029/2023MS003846.
- [A.11] L. Johnson^{*}, B. Fox-Kemper, Q. Li, H. Pham, S. Sarkar, A finite-time ensemble method for mixed layer model comparison, *Journal of Physical Oceanography* 53 (2023) 2211–2230. doi:10.1175/JPO-D-22-0107.1.

- [A.12] H. Pham^{*}, S. Sarkar, L. Johnson, B. Fox-Kemper, P. Sullivan, Q. Li, Multi-scale variability of turbulent mixing during a monsoon intraseasonal oscillation in the Bay of Bengal: an LES study, *Journal of Geophysical Research: Oceans* 128 (2023) e2022JC018959. doi:10.1029/2022JC018959.
- [A.13] J.-C. Golaz^{*}, L. P. Van Roekel, X. Zheng, A. F. Roberts, J. D. Wolfe, W. Lin, A. M. Bradley, Q. Tang, M. E. Maltrud, R. M. Forsyth, C. Zhang, T. Zhou, K. Zhang, C. S. Zender, M. Wu, H. Wang, A. K. Turner, B. Singh, J. H. Richter, Y. Qin, M. R. Petersen, A. Mametjanov, P.-L. Ma, V. E. Larson, J. Krishna, N. D. Keen, N. Jeffery, E. C. Hunke, W. M. Hannah, O. Guba, B. M. Griffin, Y. Feng, D. Engwirda, A. V. Di Vittorio, C. Dang, L. M. Conlon, C.-C.-J. Chen, M. A. Brunke, G. Bisht, J. J. Benedict, X. S. Asay-Davis, Y. Zhang, X. Zeng, S. Xie, P. J. Wolfram, T. Vo, M. Veneziani, T. K. Tesfa, S. Sreepathi, A. G. Salinger, M. J. Prather, S. Mahajan, Q. Li, P. W. Jones, R. L. Jacob, J. E. J. R. Eyre, G. W. Huebler, X. Huang, B. R. Hillman, B. E. Harrop, J. G. Foucar, Y. Fang, D. S. Comeau, P. M. Caldwell, T. Bartoletti, K. Balaguru, M. A. Taylor, R. B. McCoy, L. R. Leung, D. C. Bader, The DOE E3SM model version 2: Overview of the physical model, *Journal of Advances in Modeling Earth Systems* 14 (2022) e2022MS003156. doi:10.1029/2022MS003156.
- [A.14] C. Zhu^{*}, J. Zhang^{*}, Z. Liu, B. Otto-Bliesner, C. He, E. Brady, R. Tomas, Q. Wen, Q. Li, C. Zhu, S. Zhang, L. Wu, Antarctic warming during Heinrich Stadial 1 in a transient isotope-enabled deglacial simulation, *Journal of Climate* 35 (2022) 3753–3765. doi:10.1175/JCLI-D-22-0094.1.
- [A.15] H. Wang, C. Dong^{*}, B. Fox-Kemper^{*}, Q. Li, Y. Yang, X. Chen, K. T. Lim Kam Sian, Parameterization of ocean surface wave-induced mixing using large eddy simulations (LES) II, *Deep Sea Research Part II: Topical Studies in Oceanography* 203 (2022) 105167. doi:10.1016/j.dsr2.2022.105167.
- [A.16] X. Zheng^{*}, Q. Li, T. Zhou, Q. Tang, L. Van Roekel, J.-C. Golaz, Description of historical and future projection simulations by the global coupled E3SMv1.0 model as used in CMIP6, *Geoscientific Model Development* 15 (9) (2022) 3941–3967. doi:10.5194/gmd-15-3941-2022.
- [A.17] P. Orenstein, B. Fox-Kemper^{*}, L. Johnson, Q. Li, A. Sane, Evaluating coupled climate model parameterizations via skill at reproducing the monsoon intraseasonal oscillation, *Journal of Climate* 35 (6) (2022) 1873–1884. doi:10.1175/JCLI-D-21-0337.1.
- [A.18] Q. Li^{*}, J. Bruggeman, H. Burchard, K. Klingbeil, L. Umlauf, K. Bolding, Integrating CVMix into GOTM (v6.0): A consistent framework for testing, comparing, and applying ocean mixing schemes, *Geoscientific Model Development* 14 (7) (2021) 4261–4282. doi:10.5194/gmd-14-4261-2021.
- [A.19] Q. Li^{*}, L. Van Roekel, Towards multiscale modeling of ocean surface turbulent mixing using coupled MPAS-Ocean v6.3 and PALM v5.0, *Geoscientific Model Development* 14 (4) (2021) 2011–2028. doi:10.5194/gmd-14-2011-2021.
- [A.20] Q. Li^{*}, B. Fox-Kemper, Anisotropy of Langmuir turbulence and the Langmuir-enhanced mixed layer entrainment, *Physical Review Fluids* 5 (1) (2020) 013803. doi:10.1103/PhysRevFluids.5.013803.
- [A.21] P. M. Caldwell^{*}, A. Mametjanov, Q. Tang, L. P. Van Roekel, J.-C. Golaz, W. Lin, D. C. Bader, N. D. Keen, Y. Feng, R. Jacob, M. E. Maltrud, A. F. Roberts, M. A. Taylor, M. Veneziani, H. Wang, J. D. Wolfe, K. Balaguru, P. Cameron-Smith, L. Dong, S. A. Klein, L. R. Leung, H.-Y. Li, Q. Li, X. Liu, R. B. Neale, M. Pinheiro, Y. Qian, P. A. Ullrich, S. Xie, Y. Yang, Y. Zhang, K. Zhang, T. Zhou, The DOE E3SM coupled model version 1: Description and results at high resolution, *Journal of Advances in Modeling Earth Systems* 11 (12) (2019) 4095–4146. doi:10.1029/2019MS001870.
- [A.22] Q. Li^{*}, B. G. Reichl, B. Fox-Kemper, A. Adcroft, S. Belcher, G. Danabasoglu, A. Grant, S. M. Griffies, R. W. Hallberg, T. Hara, R. Harcourt, T. Kukulka, W. G. Large, J. C. McWilliams, B. Pearson, P. Sullivan, L. Van Roekel, P. Wang, Z. Zheng, Comparing ocean surface boundary vertical mixing schemes including Langmuir turbulence, *Journal of Advances in Modeling Earth Systems* 11 (11) (2019) 3545–3592. doi:10.1029/2019MS001810.

- [A.23] B. G. Reichl*, Q. Li, A parameterization with a constrained potential energy conversion rate of vertical mixing due to Langmuir turbulence, *Journal of Physical Oceanography* 49 (11) (2019) 2935–2959. doi:10.1175/JPO-D-18-0258.1.
- [A.24] A. B. Villas Boas*, F. Ardhuin, A. Ayet, M. A. Bourassa, B. Chapron, P. Brandt, B. D. Cornuelle, J. T. Farrar, M. R. Fewings, B. Fox-Kemper, S. T. Gille, C. Gommenginger, P. Heimbach, M. C. Hell, Q. Li, M. Mazloff, S. T. Merrifield, A. Mouche, M.-H. Rio, E. Rodriguez, J. D. Shutler, A. C. Subramanian, E. J. Terrill, M. Tsamados, C. Ubelmann, E. van Sebille, Integrated observations and modeling of global winds, currents, and waves: Requirements and challenges for the next decade, *Frontiers in Marine Science* 6 (2019) 425. doi:10.3389/fmars.2019.00425.
- [A.25] Q. Li*, B. Fox-Kemper, Assessing the effects of Langmuir turbulence on the entrainment buoyancy flux in the ocean surface boundary layer, *Journal of Physical Oceanography* 47 (12) (2017) 2863–2886. doi:10.1175/JPO-D-17-0085.1.
- [A.26] Q. Li*, B. Fox-Kemper, Ø. Breivik, A. Webb, Statistical models of global Langmuir mixing, *Ocean Modelling* 113 (2017) 95–114. doi:10.1016/j.ocemod.2017.03.016.
- [A.27] Q. Li*, A. Webb, B. Fox-Kemper, A. Craig, G. Danabasoglu, W. G. Large, M. Vertenstein, Langmuir mixing effects on global climate: WAVEWATCH III in CESM, *Ocean Modelling* 103 (2016) 145–160. doi:10.1016/j.ocemod.2015.07.020.
- [A.28] H. Yang*, K. Wang, H. Dai, Y. Wang, Q. Li, Wind effect on the Atlantic meridional overturning circulation via sea ice and vertical diffusion, *Climate Dynamics* 46 (11) (2016) 3387–3403. doi:10.1007/s00382-015-2774-z.
- [A.29] H. Yang*, Y. Zhao, Z. Liu, Q. Li, F. He, Q. Zhang, Heat transport compensation in atmosphere and ocean over the past 22,000 years, *Scientific Reports* 5 (2015) 16661. doi:10.1038/srep16661.
- [A.30] H. Yang*, Q. Li, K. Wang, Y. Sun, D. Sun, Decomposing the meridional heat transport in the climate system, *Climate Dynamics* 44 (9) (2015) 2751–2768. doi:10.1007/s00382-014-2380-5.

PUBLICATIONS IN PROGRESS

- [M.1] Q. Li*, G. L. Wagner, Transition to marginal instability in Langmuir turbulence in shallow water and an analogy with rotating plane Couette flow, *Journal of Fluid Mechanics*, Under review (2026).
- [M.2] X. Chai, Y. Ma, J.-H. Xie, Q. Li, Y. Wang*, Observation-informed numerical study of the formation of diffusive-convection staircases within double-core eddies in the arctic ocean, *Journal of Geophysical Research: Oceans*, Under review (2026).

INVITED CONFERENCE PRESENTATIONS

- [I.1] Q. Li, Stabilizing wave-induced stratification and its effects on vertical mixing in ocean surface boundary layer, in: ICERM Hot Topics Workshop - Synthesizing Research on Ocean Surface Waves in the New Arctic, Providence, RI, USA, 2025, Talk.
- [I.2] Q. Li, Surface-wave driven mixing, in: Gordon Research Conference on Ocean Mixing, South Hadley, MA, USA, 2024, Discussion Leader.
- [I.3] Q. Li, Langmuir turbulence in a diurnal cycle, in: 6th Xiamen Symposium on Marine Environmental Sciences, Virtual Meeting Online, 2023, Talk.
- [I.4] Q. Li, Modeling the ocean surface boundary layer vertical mixing by Langmuir turbulence, in: 9th Warnemünde Turbulence Days (WTD) on Ocean Mixing and its Efficiency, Putbus, Germany, 2019, Talk.

- [I.5] Q. Li, Langmuir turbulence and its effects on global climate, in: Physical Oceanography Dissertation Symposium X, Kailua-Kona, HI, USA, 2018, Talk.

CONTRIBUTED CONFERENCE PRESENTATIONS

- [P.1] Q. Li, X. Jiang, Impact of parameterized wind-driven boundary layer turbulence on the development and evolution of submesoscale mixed layer eddies in an idealized front, in: The International Workshop on Oceanic Finescale Processes and Their Impacts on Biogeochemistry, Weather, and Climate, Sanya, Hainan, China, 2026, Talk.
- [P.2] X. Jiang, Q. Li, Wave-modified symmetric instability and its impact on momentum transport, in: Gordon Research Conference on Ocean Mixing, South Hadley, MA, USA, 2026, Poster.
- [P.3] X. Jiang, Q. Li, Impact of parameterized wind-driven boundary layer turbulence on the evolution of submesoscale mixed layer eddies, in: Gordon Research Seminar on Ocean Mixing, South Hadley, MA, USA, 2026, Poster.
- [P.4] Q. Li, G. L. Wagner, Transition to marginal instability in Langmuir turbulence in shallow water and an analogy with rotating plane Couette flow, in: Gordon Research Conference on Ocean Mixing, South Hadley, MA, USA, 2026, Poster.
- [P.5] Q. Li, Large eddy simulations of stabilizing wave-induced stratification and effects of the sub-grid scale closure, in: Ocean Sciences Meeting, Glasgow, Scotland, 2026, Poster.
- [P.6] Z. Xu, L. Yu, Q. Li, Z. Chen, Can trait-based insights improve fixed-type models? An intercomparison for climate and disturbance projections, in: Ocean Sciences Meeting, Glasgow, Scotland, 2026, Poster.
- [P.7] Q. Li, Large eddy simulations of stabilizing effects induced by opposing Eulerian current and Stokes drift in an idealized ocean surface boundary layer, in: 12th Warnemünde Turbulence Days (WTD) on Waves and Turbulence, Putbus, Germany, 2025, Talk.
- [P.8] X. Jiang, Q. Li, Interaction between boundary layer turbulence and submesoscale mixed layer eddies and its influence on upper ocean stratification, in: European Geosciences Union (EGU) General Assembly, Vienna, Austria, 2025, Poster.
- [P.9] S. Zou, Q. Li, Fuse the SAR and SWIM observations to acquire global Stokes drift estimations, in: European Geosciences Union (EGU) General Assembly, Vienna, Austria, 2025, Poster.
- [P.10] J. Huang, M. Chamecki, Q. Li, B. Chen, The role of diurnal forcings on the merging of surface and bottom boundary layers in the coastal ocean and their implications on vertical mixing, in: 7th Xiamen Symposium on Marine Environmental Sciences, Xiamen, Fujian, China, 2025, Poster.
- [P.11] X. Jiang, Q. Li, Influence of boundary layer turbulence on submesoscale mixed layer eddy-induced restratification, in: 7th Xiamen Symposium on Marine Environmental Sciences, Xiamen, Fujian, China, 2025, Poster.
- [P.12] W. Pan, Q. Li, Transient response of Langmuir turbulence structure to abrupt changes in surface buoyancy forcing, in: 7th Xiamen Symposium on Marine Environmental Sciences, Xiamen, Fujian, China, 2025, Poster.
- [P.13] Z. Wei, Q. Li, B. Chen, A direct comparison of turbulent fluxes in ocean boundary layer vertical mixing parameterizations, in: 7th Xiamen Symposium on Marine Environmental Sciences, Xiamen, Fujian, China, 2025, Poster.
- [P.14] Z. Xu, L. Yu, Q. Li, Episodic phytoplankton blooms in the low-latitude North Atlantic: Causes and contributions to annual carbon sequestration, in: 7th Xiamen Symposium on Marine Environmental Sciences, Xiamen, Fujian, China, 2025, Talk.

- [P.15] S. Zou, Q. Li, A transformer based approach for fusing SAR and SWIM measurement to produce better-quality wave spectrum, in: 7th Xiamen Symposium on Marine Environmental Sciences, Xiamen, Fujian, China, 2025, Poster.
- [P.16] W. Pan, Q. Li, Assessing the impact of transient forcing on Langmuir turbulence intensity and the resulting vertical mixing in the upper ocean, in: AGU Annual Meeting, Washington, D.C., USA, 2024, Poster.
- [P.17] Z. Wei, Q. Li, B. Chen, Assessment of vertical mixing parameterizations in modelling merging surface and bottom boundary layers, in: AGU Annual Meeting, Washington, D.C., USA, 2024, Poster.
- [P.18] Q. Li, Z. Wei, B. Chen, A direct assessment of Langmuir turbulence parameterizations in idealized merging boundary layers in coastal oceans, in: 3rd Hong Kong and Macau Ocean Forum & Areas of Excellence (AoE) Forum, Hong Kong, China, 2024, Talk.
- [P.19] Q. Li, A hybrid mass-flux/high-order turbulence closure for ocean vertical mixing, in: 3rd Youth Forum on Marine Science, Shanghai, China, 2024, Talk.
- [P.20] Q. Li, W. Pan, Transient response of Langmuir turbulence in a diurnal cycle, in: Asia Oceania Geosciences Society 21th Annual Meeting, Pyeongchang, Gangwon-do, South Korea, 2024, Poster.
- [P.21] Q. Li, Z. Wei, B. Chen, Assessing the performance of Langmuir turbulence parameterizations in idealized merging boundary layers in coastal oceans, in: Ocean Sciences Meeting, AGU/ASLO/TOS, New Orleans, LA, USA, 2024, Poster.
- [P.22] Q. Li, Z. Wei, B. Chen, Assessing the performance of Langmuir turbulence parameterizations in idealized estuarine-shelf environment, in: AGU Annual Meeting, AGU, San Francisco, CA, USA, 2023, Talk.
- [P.23] Q. Li, Z. Wei, B. Chen, Towards a better parameterization of ocean turbulent mixing in the estuarine-shelf environment, in: Asia Oceania Geosciences Society 20th Annual Meeting, Singapore, 2023, Talk.
- [P.24] W. Pan, Q. Li, Transient response of Langmuir turbulence to the changing forcings in a diurnal cycle, in: Asia Oceania Geosciences Society 20th Annual Meeting, Singapore, 2023, Poster.
- [P.25] Z. Wei, Q. Li, B. Chen, Modeling the ocean boundary layer turbulent mixing in the transitioning regions between coastal and open oceans, in: Asia Oceania Geosciences Society 20th Annual Meeting, Singapore, 2023, Poster.
- [P.26] Z. Xu, L. Yu, Q. Li, Impact of phytoplankton light absorption on upper ocean dynamics and carbon export production in the subpolar North Atlantic: A modeling assessment, in: Asia Oceania Geosciences Society 20th Annual Meeting, Singapore, 2023, Poster.
- [P.27] Q. Li, Z. Wei, B. Chen, Progress towards modeling the ocean boundary layer turbulence from open oceans to coastal oceans, in: 2nd Ocean Forum in Hong Kong and Macao, Hong Kong, China, 2023, Talk.
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